

4. Bubble Sort :- bagal wale se compare karo.

arr =

10	1	7	6	14	9
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↖ swap ↗

Round 1:-

$\left\{ \begin{array}{l} 10 \ 7 \ 6 \ 14 \ 9 \\ 7 \ 10 \ 6 \ 14 \ 9 \\ 7 \ 6 \ 10 \ 14 \ 9 \\ 7 \ 6 \ 10 \ 14 \ 9 \\ 7 \ 6 \ 10 \ 9 \ 14 \end{array} \right.$
↖ swap ↗ 1st → (n-1)
↖ swap ↗
↖ swap ↗
↖ swap ↗
↖ swap ↗ → largest element in the end.

Round 2:-

$\left\{ \begin{array}{l} 7 \ 6 \ 10 \ 9 \ 14 \\ 6 \ 7 \ 10 \ 9 \ 14 \\ 6 \ 7 \ 10 \ 9 \ 14 \\ 6 \ 7 \ 9 \ 10 \ 14 \\ 6 \ 7 \ 9 \ 10 \ 14 \end{array} \right.$
↖ swap ↗ 2nd → (n-2)
↖ swap ↗
↖ swap ↗
↖ swap ↗
↖ swap ↗ → 2nd largest element sorted hogaya.

Round 3:-

$\{$
 1 6 7 9 10 14
 $\searrow$
 1 6 7 9 10 14
 $\searrow$
 1 6 7 9 10 14
 $\searrow$
 1 6 7 9 10 14

2nd $\rightarrow (n-3)$

Third largest element
sort ho gaya.

Round 4:-

$\{$
 1 6 7 9 10 14
 $\searrow$
 1 6 7 9 10 14
 $\searrow$
 1 6 7 9 10 14

3rd $\rightarrow (n-4)$

Fourth largest element
sort ho gaya.

Round 5:-

$\{$
 1 6 7 9 10 14
 $\searrow$
 1 6 7 9 10 14

Fifth largest element
sort ho gaya.

$\therefore$ largest element go bahega wo apne
ap hi shi position pe chala jayega.

```

for (i = n-2; i >= 0; i--) {
    for (j = 0; j <= i; j++) {
        if (arr[j] > arr[j+1]) {
            swap(arr[j], arr[j+1])
        }
    }
}
  
```